Knee MRI Abnormality Detection

Model specification: RSNA-style multimodal knee MRI challenge (12 findings, macro-AUC)

1. Objective

Rank abnormal knees above normal knees for each of 12 findings (ACL, MCL, Medial Meniscus, Lateral Meniscus, Medial OA, Lateral OA, PF OA, Effusion, Synovitis, Baker's, Contusion, Fracture), on scans from sites never seen in training.

- **Metric:** macro-averaged ROC AUC across the 12 labels. Only score order matters; thresholds and calibration do not affect the score.
- **Equal label weight:** rare labels (e.g. Fracture, MCL) count as much as common ones, so the weakest labels are where the easy gains are.
- **Pooled scoring:** per-site score shifts hurt the pooled ranking, so site robustness matters.
- **Real bottlenecks:** few gold labels, noisy report-derived labels, scanner and site shift, and noisy AUC on rare labels.

2. Strategy

- Audit the data: gold-labeled vs total studies, per-label prevalence, report languages, series per study, scanner/site proxies from DICOM tags.
- Mine labels for unlabeled studies from the multilingual reports (used for training only; reports are absent at test time).
- Train an image-only, multi-series model on gold plus pseudo-labels.
- Validate only on gold labels, grouped by site, then ensemble.
- Check competition rules on external weights, LLM use, internet access and inference time limits.

3. Label mining from reports

Component	Spec
Labeler A	Multilingual encoder (XLM-R-base or mDeBERTa-v3) fine-tuned on the gold subset, 12 sigmoid outputs
Labeler B	Few-shot LLM extraction of the 12 findings with negation and hedging flags
Combination	Average A and B probabilities; keep soft labels
Trust gate	Measure each labeler against gold per label (AUC, F1). Keep pseudo-labels where AUC is about 0.9 or higher; down-weight the rest
Known traps	Negation, partial vs complete tears, post-surgical knees, hedged wording, translation differences

4. Preprocessing

- **Reading:** pydicom with gdcm or pylibjpeg (JPEG Lossless, JPEG 2000). Apply rescale slope and intercept.
- **Slice order:** project ImagePositionPatient on the plane normal; fall back to InstanceNumber.
- **Normalization:** clip to the 1st to 99.5th percentile per series, then z-score.
- **Resampling:** common in-plane spacing (about 0.6 mm), crop or pad to 224 x 224, uniformly sample 32 slices per series.
- **Caching:** store as uint8 or float16 .npy for fast epochs.
- **No horizontal flips:** medial/lateral and anterior/posterior labels are side-dependent.

5. Model

Part	Spec
Input	Up to 3 planes per study; best fluid-sensitive, fat-suppressed series per plane, plus a second series per plane when available
Backbone	Shared ImageNet-pretrained 2D encoder (ConvNeXt-Tiny or EfficientNetV2-S), 2.5D input (3 adjacent slices as channels)
Slice pooling	Attention pooling within each series
Series fusion	Learned embeddings for plane, Fluid_Sensitive and Fat_Suppression, then a 2-layer transformer across series
Head	12 sigmoid outputs
Robustness	Series dropout (p = 0.2) with a mask token for missing planes
Second architecture	3D ResNet-18 or R(2+1)D on sagittal and coronal series, for ensemble diversity
Auxiliary (train only)	Severity/location/laterality targets from reports; optional CLIP-style alignment between study embedding and report embedding

6. Loss

```
L = sum over labels of w_i * BCE(logit_i, target_i) + 0.1 * L_aux
```

- w = 1.0 for gold labels; w = 0.3 to 0.5 times labeler confidence for pseudo-labels.
- Stage 1: train on gold plus pseudo-labels. Stage 2: fine-tune on gold only for 2 to 3 epochs at low LR.
- Imbalance: mild pos_weight = sqrt(neg/pos). Heavy reweighting is unnecessary because the metric is rank-based.
- Label smoothing 0.05. Optional final stage: pairwise AUC-surrogate loss for rare labels.

7. Hyperparameters (starting point)

Parameter	Value
Optimizer	AdamW, weight decay 0.05
Learning rate	Backbone 5e-5; transformer and heads 3e-4
Schedule	1 epoch warmup, then cosine decay
Batch	6 to 8 studies with gradient accumulation, AMP, gradient checkpointing
Epochs	15 to 25 (pseudo stage), 3 (gold fine-tune)
Regularization	Drop-path 0.1, dropout 0.2, EMA 0.999
Augmentation	Rotation +/-10 deg, scale 0.85 to 1.15, translation up to 8%, gamma/contrast jitter, Gaussian noise, slice-index jitter
Tuning	15 to 20 Optuna trials at reduced resolution (LR, pseudo-label weight, number of series)

8. Validation and metrics

- **Primary:** macro-AUC on gold-labeled out-of-fold predictions.
- **Split:** 5-fold, stratified per label where possible, grouped by site proxy (manufacturer, model, field strength, institution tags). Hold out one or two sites entirely as a proxy for the hidden test.
- **Also report:** per-label AUC with bootstrap 95% CI, AUC per site and field strength, and text-labeler quality against gold.

- **Leaderboard caution:** prevalence differs between train, public and private sets. Trust CV plus the site holdout over small leaderboard gains.

9. Thresholds and calibration

- **Thresholds:** not needed for AUC.
- **Calibration:** only matters through cross-site score shifts. Reduce them with site-robust augmentation, intensity normalization, and optional per-site rank normalization if site is inferable.
- **Ensembling:** average logits or ranks across models.
- **If a clinical version is wanted later:** temperature scaling on gold OOF, per-label thresholds (sensitivity-first for triage, Youden J for balanced).

10. Ensemble and inference

- 5 folds x 2 architectures with light TTA (slice-offset shifts, no flips).
- Fallback for unreadable or missing series: output the training prior for each label, not 0.5.
- About 1,300 test studies: time 100 studies first to confirm the pipeline fits the notebook limit.
- Submission columns must match exactly (including Baker's), probabilities in [0, 1].

11. Plan for the remaining 24 days

Days	Task
1 to 3	Data audit, DICOM loader, cache, single-fold gold-only baseline
4 to 8	Report labelers, pseudo-labels, trust gating
9 to 14	Multi-series model with pseudo+gold training and auxiliary losses
15 to 19	Second architecture, site-holdout analysis, tuning
20 to 22	Full folds, ensemble, inference timing test
23 to 24	Final submission and sanity checks; keep spare submissions

12. Evidence from the supplied papers

Paper	Relevance
Vaz et al. 2005 (300 patients, arthroscopy reference)	Human benchmark: medial meniscus sens/spec 97.5/92.9; lateral meniscus 91.9/93.6; ACL 99.0/95.9; cartilage 76.1/94.9
Pedoia and Majumdar	3D CNN on 1,487 exams reached meniscus sens 89.8 / spec 82.0; warns about label noise and cross-scanner generalization
Cheng et al. meta-analysis	1.5T vs 3T similar for ligament and meniscus; 3T better for cartilage, so stratify by field strength
Kumm et al. 2018	Cartilage damage and bone marrow lesions are common even in radiographically normal knees; label definitions matter
Byrne et al. (topological loss)	Cardiac MR segmentation; relevant only if a segmentation stage is added